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[23EE31]

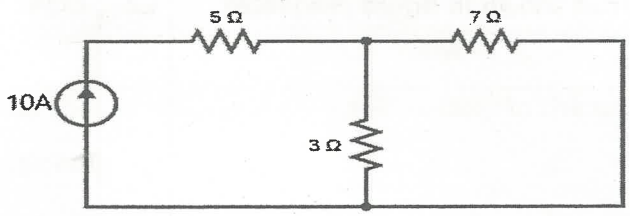
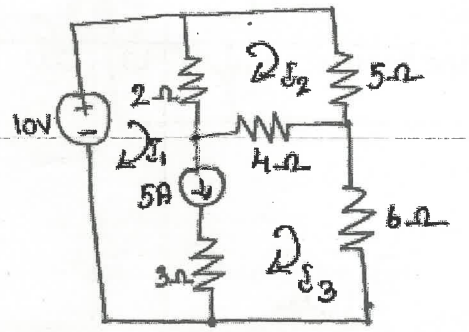
THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

ELECTRICAL CIRCUIT ANALYSIS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Compare Nortons and Thevenin's theorem.	L3	CO1	**02**
	b.	Explain Delay time and Rise time with respect to RLC series circuit with DC excitation.	L3	CO2	**02**
	c.	Express the passive elements in terms of Impedance.	L3	CO3	**02**
	d.	Write the Laplace transforms of the three singularity functions used for circuit analysis.	L4	CO5	**02**
	e.	Express Z parameters in terms of h parameters.	L4	CO6	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	For the below circuit, find the change of current, if 3Ω resistor is replaced by a 7Ω resistor using Compensation theorem. 	L3	CO1	**05**
		OR			
	b.	Find the currents through the network using mesh analysis. 	L3	CO1	**05**
3.	**a.**	Derive the roots of a RLC series circuit when excited by a DC source by assuming zero initial conditions and the switch is closed at time t=0.	L3	CO2	**05**
		OR			

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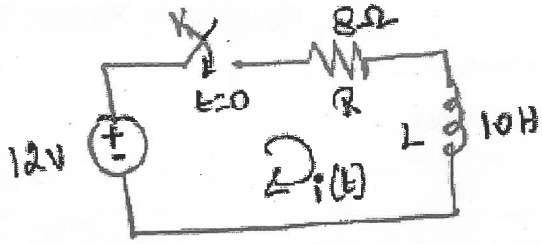
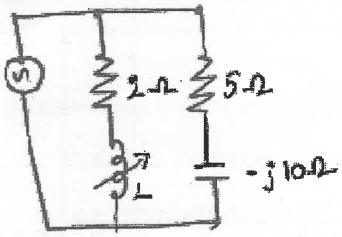
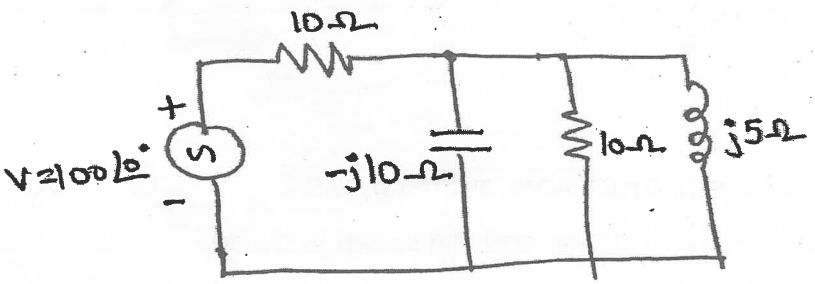
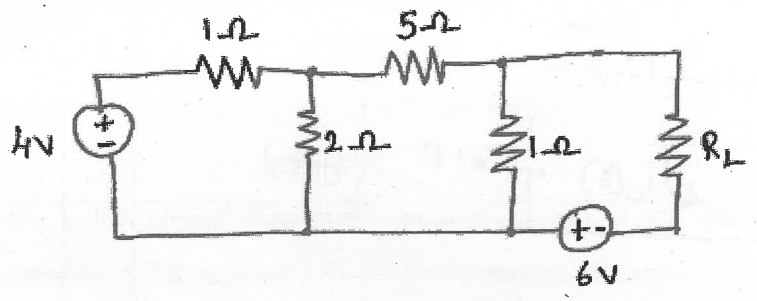
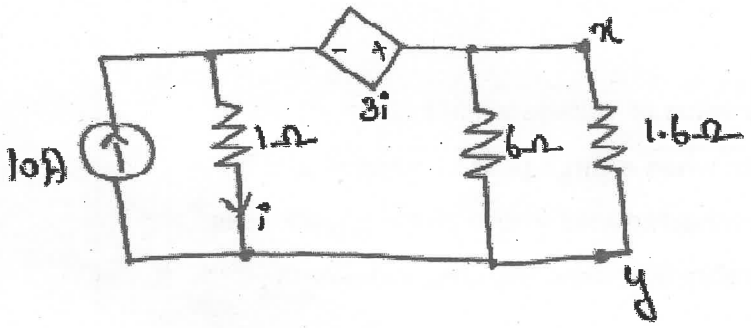
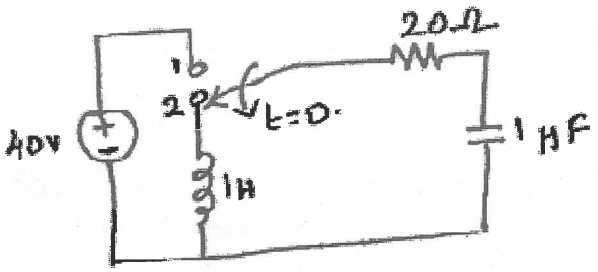
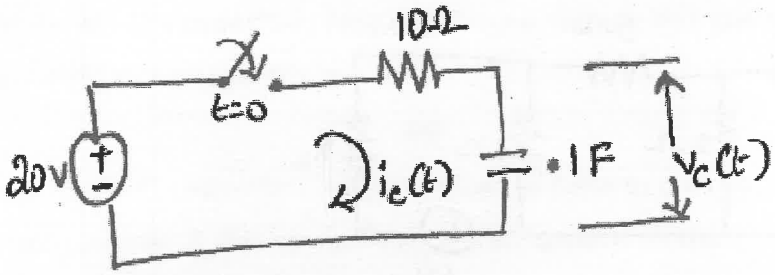
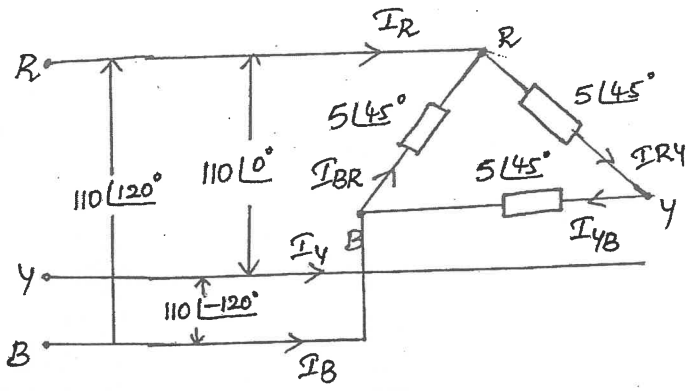
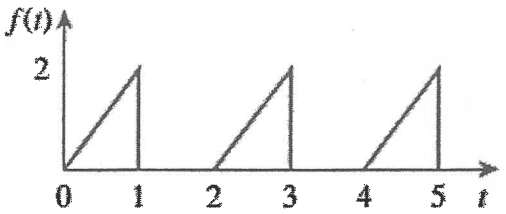
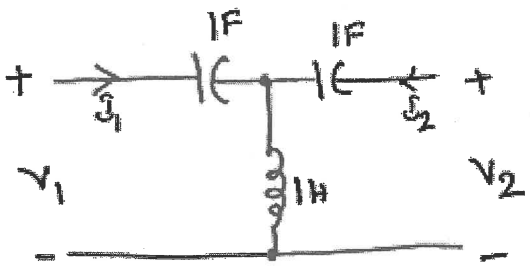
	b.	For the circuit shown, initially the switch is kept open for a long time. At t=0, Switch is closed. Find the value of current at t=0.25 sec. 	L3	CO2	**05**
4.	**a.**	A capacitor which has an internal resistance of 10Ω and a capacitance value of 100uF is connected to a supply voltage given as V(t) = 100 sin (314t). Calculate the peak instantaneous current flowing into the capacitor. Also construct a voltage triangle showing the individual voltage drops.	L3	CO3	**05**
		OR			
	b.	Three identical capacitors are connected in delta to a 415V, 50 Hz, 3 phase supply. If the line current is 15A, determine the capacitance of each capacitor.	L3	CO4	**05**
5.	**a.**	Find the value of L for which the circuit shown in figure resonates at ω_0 = 500 rad/sec. 	L2	CO4	**05**
		OR			
	b.	Find the real power delivered by the source in the figure shown. 	L3	CO3	**05**

6.	a.	Construct a circuit that realizes the following Z parameters $[Z] = \begin{bmatrix} 12 & 4 \\ 4 & 8 \end{bmatrix}$	L4	CO6	**05**
		OR			
	b.	Explain the Hybrid parameters with equivalent circuit.	L4	CO6	**05**
7.	a.	Calculate $v(t)$ given that $V(s) = \frac{s+1}{s(s^2+4s+4)}$	L4	CO5	**05**
		OR			
	b.	State Thevenins theorem and explain the steps involved in solving the problems using Thevenins theorem.	L4	CO1	**05**
		PART-C		**6 X 10=60**	
8.	a.	Find the value of R for maximum power transfer in the circuit. Calculate the power through the load. 	L3	CO1	**10**
		OR			
	b.	Find the current through 1.6Ω resistor in the circuit. Use Norton's theorem. 	L3	CO1	**10**
9.	a.	For the network shown, the switch is moved from position 1 to position 2 at t=0, the steady state has been reached before switching. Calculate i, $\frac{di}{dt}$ and d^2i/dt^2 at $t = 0^+$	L3	CO2	**10**

					
		OR			
	b.	A Series RC circuit consists of R=10Ω and c=0.1F. Initially Switch is kept open for long time. At t=0 it is closed. Find expression for $V_c(t)$, $i_c(t)$ and $V_R(t)$ against time. 	L3	CO2	**10**
10.	**a.**	Estimate the relation of current and voltage with frequency in series resonance. Also derive expression for Quality factor.	L3	CO3	**10**
		OR			
	b.	An alternating voltage is given by V=282.8 sin 314t volts. Find **i.** The rms voltage **ii.** The frequency and **iii.** The instantaneous value of voltage when t=4ms.	L3	CO3	**10**
11.	**a.**	Determine the line currents when a three phase balanced system supplies 110V to a delta connected load whose phase impedances are equal to (3.54+j3.54) ohm and draw the phasor diagram.	L2	CO4	10

					
		OR			
	b.	A voltage v(t) = 150sin 10^3t is applied to a series RLC circuit where, R=40Ω, L=0.13H and C=10μF. Find **i.** The power supplied by the source **ii.** The reactive power supplied by the source **iii.** The reactive power of the capacitor **iv.** The reactive power of the inductor **v.** The power factor of the circuit.	L2	CO4	**10**
12.	**a.**	Explain any four properties of Laplace transforms.	L3	CO5	**10**
		OR			
	b.	Calculate the Laplace transform of the periodic function in Fig. 	L3	CO5	**10**
13.	**a.**	Define h and t parameters and obtain the relationship between them. Draw the equivalent circuit of h – parameters.	L3	CO6	**10**
		OR			
	b.	Find H parameters for the circuit shown in figure. 	L3	CO6	**10**